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Supply Runner

Game Design Document – Group 4

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Design History

This is a brief description of the different game design changes and versions we have gone through to create this game. Along with the steps and different thought processes we may have had when making a change to the design of our game.

Version 1.0.

This is the current version of our game design document. We will be updating this page as we continue to work on our project and if we decide to make small changes to our game design. We will make new entries on this page when changes are made.

Section 1 - Game Overview

1.1. Concept

- The end goal of “Supply Runner” is to create a fun to play, single player zombie survival role playing game. Where the player is a survivor during a zombie apocalypse and the main goal is to fight and survive. Additionally, the player's objective is also to do jobs and missions for other survivors in need and to save them from the apocalypse as well.

1.1.1 Core Message

- You may fight alone but you are always fighting for a bigger purpose greater than yourself, for the survival of the new world of the living.
- Survive for yourself and for the lives of others counting on the supplies you find.

1.2. Feature Set

- Loot abandoned buildings and clear streets of enemies.
- Combat system that allows the player to fight enemy combatants.
- Safe areas from the open land full of zombies, where you can buy and sell supplies.
- A weapon modification system that allows for dmg mods and other specs on weapons
- Crafting system to make ammo and other supplies that are used for quests.
- Player Character Progression System.
 - o This includes knowledge skills that make killing zombies easier.

1.3. Genre

- Supply Runner is an Action, Survival, Shooter, Role Playing Game.
- Our game is an RPG because it features a leveling system where players earn XP for defeating zombies and completing quests, allowing them to upgrade their skills and attributes. The quests range from gathering resources to strategic zombie clear-outs, creating a sense of adventure and strategy. Additionally, our game's focus on trading materials as a form of currency and meaningful player choices in dialogues and actions define its RPG nature.

1.4. Target Audience

- 12 years and up.
- Players who enjoy survival games
- Players who enjoy first person shooters

1.5. Game Flow Summary

- Once a player opens our game and it's finished loading, these 4 options are displayed.
 - o Continue
 - To continue playing the most recent save file.
 - o Load
 - Access previous playthroughs
 - o New
 - Create a new game file.
 - o End / Quit
 - To fully exit the application
- The player then would be playing the game.
- When the player is done

- The player will have the option to save via the pause menu and be able to exit to main or exit the game entirely.

1.6. Look and Feel

- Post-apocalyptic environment, where there is overgrowth in vegetation. There is a lot of darkness, eeriness, fog, and cloudiness in the sky. The scene also consists of destroyed buildings, structures, fire outbreaks, oil spills, and debris.

1.7. Project Scope

1.7.1. Number of Locations

- “The Settlement”
 - This will be the main area where the player will be able to trade with NPC merchants.
 - This will be the player’s base/home where quests and activities are given to the player.
- “The Horde”
 - This will be a place where at the end of the demo the player will be given a quest to venture to a large area completely overrun with zombies and the goal is to clear it out to get access to the crucial supplies stored there.

1.7.2. Number of NPC’s

- Merchants: 3
- Zombies: The number of zombies is based on difficulty and based on the state of the game for the player.
- Survivors (NPCs that the player must save): Between 5-10, based on difficulty.

1.7.3. Number of Weapons

- Basic Assault Rifle
 - 3-6 Assault rifles some basic and some rare but subject to change.
- Shotgun
 - 1-2 shot gun options one basic and one rare.
- Pistol
 - 2-3 types of handgun options some basic some rare.
- SMG
 - 2-3 some basic some rare.

Section 2 – Gameplay and Mechanics

2.1. Gameplay

2.1.1. Game Progression

- Exp Level based system
 - o Certain amount of exp rewarded per enemy killed.
 - o Exp rewards for bringing NPCs back to the settlement.
- Player gains a point to spend in a certain skill or attribute to improve.
- The Player would gain a skill point after leveling up.
 - o The player can spend these points to improve different character attributes such as overall health, health regeneration, damage reduction.

2.1.2. Mission / Challenge Structure

- Supply Gathering missions.
 - o Missions send the player out to gather a large amount of a certain type of supplies.
- Going out get a specific item mission.
 - o Where the player would be asked by an NPC to go out and search for a specific item or craft a specific item for that NPC

2.1.3. Objectives

- “Clearing” Objectives
 - o Clearing as in taking out all the zombies in a certain area to regain control.
- The Horde
 - o There will be at least one area in our game where there is a large structure that is completely overrun with zombies that continues to accumulate with zombies, this “Nest”/” Horde” needs to be taken care of to continue the safety of the settlement.

2.2. Mechanics

5.2.1. Physics

- This Game will be implemented using a normal sense of gravity, as to be realistic to the real world.
- The player will be able to collide and run into any visible objects as expected and the game should feel like the real world in the aspect of gravity and interacting with objects.

5.2.1.2 Metrics / Units

- Gravity
 - o -9.81m per sec^2
 - o The player will move at a speed of $7.0\text{ units per second}^2$
 - As well as different drag values when the player is on different surfaces.

2.2.1. Movement

2.2.2.1. General Movement

- o The player will be able to move in all cardinal directions.

- The player will be able to climb ladders and other visibly climbable objects.
- The player will be able to have a controlled jump.
 - There won't be any double jumping as we are trying to make the movement as realistic as possible.
- The player will also be able to crouch down to hide (subject to change)

2.2.3. Objects

5.2.3.1. Picking Up Objects

- The Player will be able to pick up floating objects/items dropped by enemies.
- The Player will also be able to open/interact with chests and storage boxes when looting.

2.2.4. Actions

5.2.4.1. Switches and Buttons

- The player will be able to access and interact with the world around them with designated interaction items such as doors, switches, and buttons.
 - This will occur when the user presses the assigned interact key.

2.2.4.2. Picking Up, Carrying and Dropping

- Picking up
 - Items will be picked up when the player either interacts with a chest or storage crate and takes form the UI.
 - Items like drops will be picked up from the ground, the items will probably be floating in place.
- Carrying
 - the player will carry the items gathered in their inventory and be able to craft and manipulate items through the inventory screen.
- Dropping
 - The player won't be able to physically drop items or objects, but the player will be able to potentially throw grenades.
 - Also, the player instead of dropping unwanted items our inventory system will allow a deletion/garbage area to get rid of items.

2.2.4.3. Talking

- Talking will only be done by NPC merchants and Players.
- They will make statements about what items they are willing to trade with the player.
- Players will be given 4 choices to choose from for what to say back to the NPC merchant.

2.2.4.4. Reading

- Speech bubbles will appear when an NPC speaks.
- 3-4 conversation response options will appear for the player, and they can decide what to say back to the NPC merchant.

2.2.5. Combat

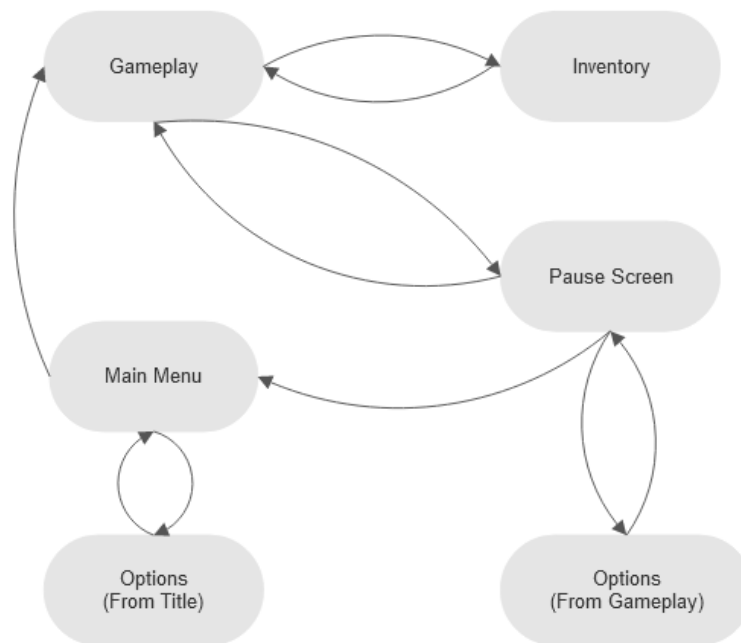
- The Combat System of our project will consist of ranged combat via kinetic modern weapons / firearms.
- Only combat between the player and zombies

2.2.6. Economy

- Economy of our game is based on trading items.
 - o Drops from zombies and or items looted in the world will be tradable to the merchants in the settlement area for other items.
- The player will also be able to complete quests/requests by giving NPC's that are asking for a specific item through trade of goods.
- Since this is a post-apocalyptic world, we decided the having a currency may not make sense for the state of the in-game world.

2.3. Screen Flow

2.3.1. Screen Flow Chart



2.3.2. Screen Descriptions

- See 5.1.1 and 5.1.2 for detailed information on the different screens.

2.4. Game Options

- The game can be configured with different options to tailor the player's experience.
- The player can configure their sensitivity, volume, and their Mini-Map mode.
 - o For more information, see the options screen in 5.1.2.3.

2.5. Replaying and Saving

- The player will be able to save and load current and previous save games.

- This will be navigated in the options menu and the main menu.

2.6. Easter Eggs

- We will just like the first game to ever have an easter egg, each of the team members' names will be placed somewhere in our games world for the player to find.

Section 3 – Story, Setting, and Character

3.1. Story and Narrative.

- You are one of few survivors left on earth, and you must survive by making supply runs to collect loot, supplies, and eliminate zombies along the way of your journey.

3.1.1. Back story

- There has been a catastrophic outbreak of a virus, and a lot of the human population has turned into zombies. The remaining few are sticking together and it's our character's job to go into zombie infested lands to clear out areas and loot supplies.

3.1.2. Plot Elements

- You find other survivors while you are collecting loot and supplies for your safe zone, and you have the choice of accepting them into your safe zone or ignoring them and continuing with your own tasks.

3.1.3. Game Progression

- The user progresses by clearing areas and streets within the city and unlocks more areas as they eliminate zombies.

3.2. Game World

3.2.1. General look and feel of world.

- Very eerie, dark, foggy, smokey, post-apocalyptic, abandoned buildings.

3.2.2. Area #1

3.2.2.1. General Description

- Safe zone for the character and has interactable NPCs to get requests such as find supplies.

3.2.2.2. Physical Characteristics

- Run down warehouse with fenced off areas, and supplies stored for remaining survivors.

3.2.2.3. Connections to other areas

- Road connected to other main city roads where the dangerous areas are.

3.2.3. Area #2

3.2.3.1. General Description

- This area is within the city, very dangerous with zombies roaming around the streets.

3.2.3.2. Physical Characteristics

- Urban area, many buildings however they are all abandoned, destroyed, or filled with zombies.

3.2.3.3. Connections to other areas

- Road connected to other main city roads where the dangerous areas are.

3.3. Characters

3.3.1. Character #1

- The Main Character (The User)
 - The user controls this character and is given decisions on how the character wants to progress.

3.3.1.1. Back story

- Family was killed by zombies; character is physically and mentally strong from working out.

3.3.1.2. Personality

- Brave, curiosity, ambitious, adventurous, charismatic, leadership

3.3.1.3. Look

3.3.1.3.1. Physical characteristics

- Adult man, dirty clothes, and unsanitary because of environmental causes.

3.3.1.3.2 Animations

- Reload animations and an animation to toss a grenade for special ability.

3.3.1.4. Special Abilities

- Grenade ability, when a certain number of zombies have been eliminated by the user an ability option is given to the user to press to the user.

3.3.1.5 Relevance to game story

- The character is important to the story because supplies are required, and it helps the survival of all the people that reside in the safe zone.

3.3.1.6 Relationship to other characters

- one of the few survivors residing in the safe warehouse, working together to live through the apocalypse.

Section 4 – Levels

7.1. Tutorial

- When the player first loads into the Map of our game, they are brought through a short tutorial of how the gameplay mechanic's work.
- A zombie will appear on the screen and the tutorial will start.
- The tutorial will show how to shoot the zombie and how to pick up the item that the zombie drops.
- It will also show how to reload the gun, how to jump, how to walk (forward, backwards, left, and right)

7.2. The Main Game

- After the player is finished with the tutorial, the real game begins, and they are now able to explore the map, shoot zombies that start spawning, interact with merchants, and save the survivors that need help in the city.

7.3. Activity - "The Horde"

- The Horde is a special quest in our game that the player can perform once they reach level 10 (XP level). This quest involves venturing to a large area completely overrun with zombies and the goal is to clear it out to get access to the crucial supplies stored there. It gives the player an opportunity to gather new supplies and or weapons that will help them advance in the game even further than before.

Section 5 – Interface

5.1. Visual System

5.1.1. The HUD

5.1.1.1 Mockup



5.1.1.2. Description

- The HUD shows all the information the player needs to be immediately aware of in the corners of the screen:

5.1.1.2.1. The Mini-Map

- Shows a map of the player's surroundings. The map is centered on the player, who is marked with a red triangle that points in the direction that they are facing.

5.1.1.2.2. The Objective

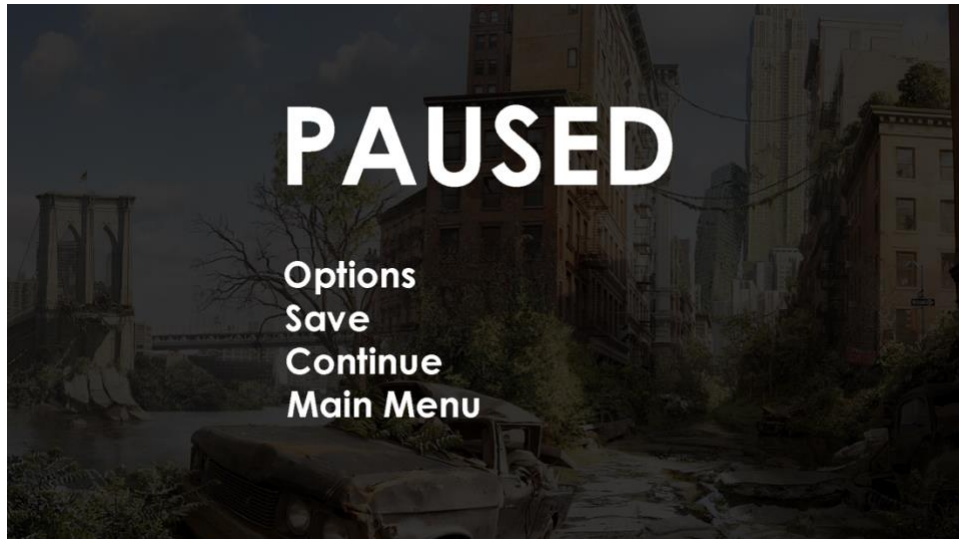
- Shows what the player needs to be currently doing, and their progress in the form of a white bar that fills up with green (if applicable).

5.1.1.2.3. Combat Information

- The red bar represents the player's health and empties and fills up in accordance.
- The grey rectangle shows information about the player character's equipped weapon.
 - On the right side of and under the ammunition, the icon and name of the player's weapon are displayed, respectively.
 - The darker number shows how much ammunition the player's weapon has loaded, and the lighter number shows the amount of ammunition the player has in their inventory.
 - The grey and yellow bar shows how much ammunition is in the player's weapon compared to how much it can hold. When there is no yellow, the player's weapon is empty.

5.1.2. Menus

5.1.2.1 The Pause Menu



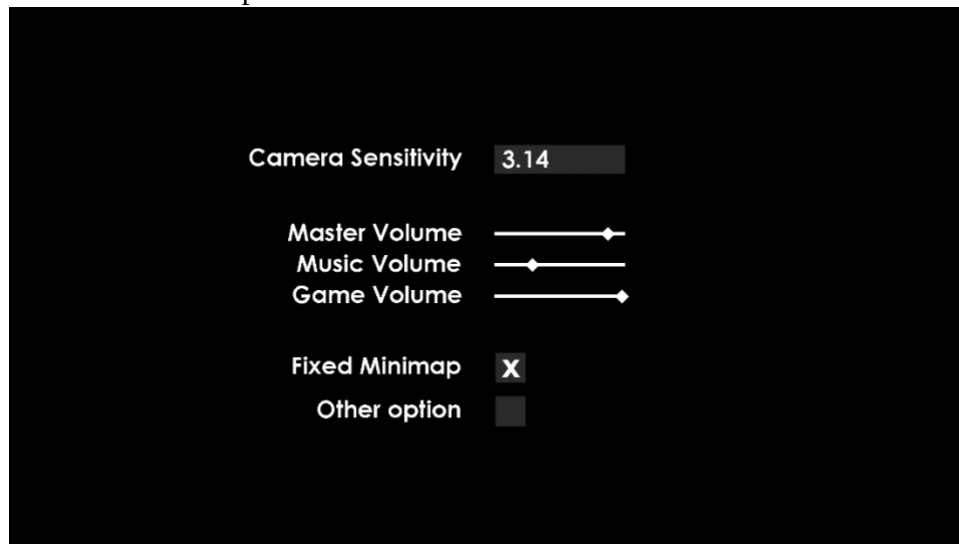
- Options: Go to the options screen and configure the game settings from there.
- Save: Save the player's progress to be loaded later.
- Continue: Exit the pause menu.
- Main Menu: Return to the main menu.

5.1.2.2. The Main Menu



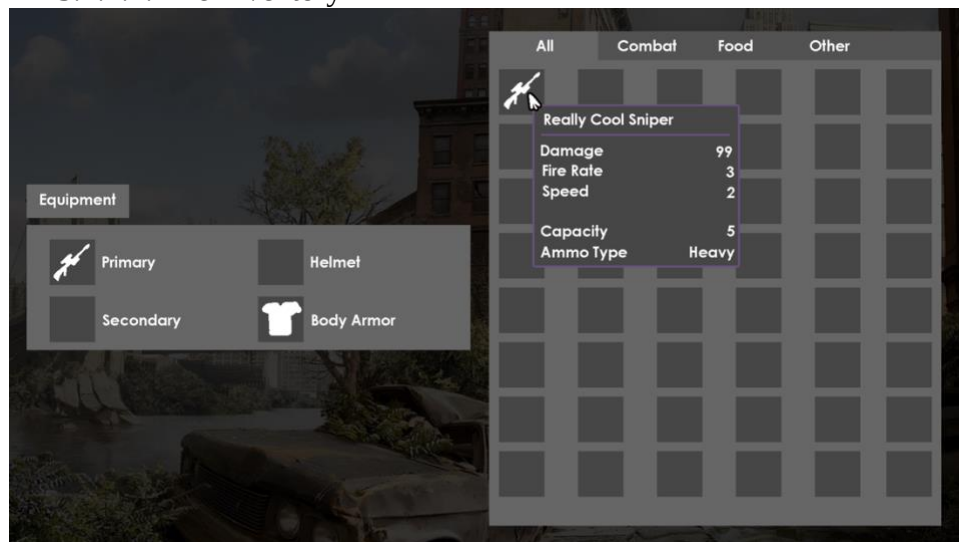
- Continue: Continue the game from the player's save data.
- New Game: Play the game from the beginning instead of from the save file.
- Options: Go to the options screen and configure the game settings from there.

5.1.2.3. The Options Menu



- Camera Sensitivity (value): Controls how fast the camera moves.
- Master Volume (slider): Controls the volume for the entire game.
- Music Volume (slider): Controls the volume for the music only.
- Game Volume (slider): Controls the volume for the game's sound effects (essentially, everything except for the music).
- Fixed Mini-Map (toggle): Whether the Mini-Map stays in place or rotates with the player.
- Other Option (toggle): Not in the final game. In the mockup to show what an unchecked toggle looks like.

5.1.2.4. The Inventory



- This is where all of the player's items (weapons, armor, ammunition, food, etc.) goes.
- Different tabs filter the inventory screen, only showing items specific to each tab.

- Hovering over an item as shown in the mockup reveals information about said item.
- On the left, the player's currently equipped items are shown.

5.1.3. Camera

- The Camera will be controlled via the position of the user's mouse in space.

5.2. Control System

5.2.1. Movement

- The Player will controller the character with (W,A,S,D) and SPACE
 - o There will also be other interactable key bindings for picking things up and other action which will be decided later.
- We will be using unity's built-in Input-System and Character Controller for modularity within the game.

5.2.2. View

- The player view/fps camera will be controlled via the mouse orientation and position to suit the mouse movement.

5.3. Audio

- Voices are given to the Player, NPC Merchants, and Survivors

5.4. Music

- Somewhat scary music will be playing in the main menu starting screen of the game.
- Adventurous/Scary music will be playing when the player is loaded into the game. This is to match the vibe of the setting of the game and current state of the abandoned post-apocalyptic city.

5.5. Sound Effects

- Gun sounds when a gun is shot.
- Stepping sounds when a player is walking.
- Zombie sounds occur when a zombie is near the player.
- Sounds occur when a player opens a door and climbs a ladder.
- Looting sound when a player picks up materials.

Section 6 – Artificial Intelligence

6.1. Opponent AI

- Once the user is within the area, the zombies will be attracted to the player and start walking towards the user. The user can easily be overwhelmed by hordes of zombies if they do not strategically move around the map.

6.2. Enemy AI

- Enemy AI are zombies, different variants (e.g. boy, girl, man, woman, sizes, health).

6.3. Non-combat Characters

- These characters consist of NPCs that give out quests and are interactable, or they could be merchants behind stalls where you can trade with them.

6.4. Support AI

6.4.1. Player and Collision Detection

- Collision events, when a zombie collides with the player such as an attack animation from a zombie.
- Kinematic and rigid bodies, to make the user and zombies not able to pass through each other, and there is a collision.

6.4.2. Pathfinding

- Waypoints, when a user receives a quest, there will be an indication to direct the user on where to go to reach the designated area. (An arrow, or how many meters away the user is from the destination)

Section 7 – Technical Bible

7.1. Target Hardware

- The Game will be designed to be played on the PC.

7.2. Development hardware and software

- Developed on Unity Game engine using C# Scripting

7.3. Development procedures and standards

- We will be using a Git Repo on GitHub to collaborate our code and resources.
- We will also be following a Feature Branch development system where every major feature of our game will be developed in its own branch for more modular testing environments.
- In Our Repo we will have three important branches
 - o Production
 - To always have a working golden standard build of our game
 - o Staging
 - For things that are ready for production to be tested extensively
 - o Development
 - And for personal testing when developing.
- The team member will branch off of staging when wanting to work on something new as Staging should be next to bug free and good to branch from.

Section 8 – References

These are the 3 games we took inspiration from while creating our game, and what makes us different from these games:

Dark and Darker

Our game draws inspiration from the gritty and intense atmosphere of "Dark and Darker." We admire its challenging dungeon levels and the high stakes of losing gear upon death, which fills each mission with a sense of accomplishment and excitement. Our game contains these nerve-wracking adventures but steps away from PvP (player vs. player) and magical spells. We're focusing on a purely PvE (player vs. environment) experience, where survival and strategy against the undead take place, and the fantasy elements are replaced with a post-apocalyptic reality.

Death Stranding

"Death Stranding" inspired us with its unique take on delivery and survival. It also inspired our focus on community and the importance of gathering and transporting supplies. But we've avoided alien narratives and extreme weather conditions, opting for a more straightforward zombie survival context that puts the emphasis on the human element and the fight for survival.

The Last of Us

We admire "The Last of Us" for its emotional storytelling and the way it brings abandoned cities to life, filled with both threats and allies. Yet, our path is different by leaving out the open-world exploration and companion dynamics. Our game is more about personal survival, offering a focused journey through structured levels where every resource counts in staying one step ahead of the zombies.

Appendices

1.1 Asset list (Subject to Change)

- City Assets
 - o <https://assetstore.unity.com/packages/3d/environments/urban/modular-lowpoly-streets-free-192094>
 - o <https://assetstore.unity.com/packages/3d/environments/urban/low-poly-street-pack-67475>
- Gun Assets
 - o <https://assetstore.unity.com/packages/3d/props/guns/low-poly-fps-weapons-lite-245929>
 - o <https://assetstore.unity.com/packages/3d/props/guns/sample-low-poly-guns-pack-207957>